

Ch 5 Worked problems

1) Total params = FFW + Attn + vocab

$$\text{FFW} = 3LDF = 8.5e9$$

$$\text{Attn} = 4LDHN = 4.2e9$$

$$\text{Vocab} = 2VD = 0.33e9$$

$$\therefore (8.5 + 4.2 + 0.33) \times 10^9 = 13.03 \times 10^9 = 13B //$$

2) 10 bytes per param for parameters and the optimizer states
 $= 130 \times 10^9 = 1306B$

for activations:

$$\text{Gate}[B, F] = \text{In}[B, D] @ W_{\text{gate}}[D, F] \Rightarrow BF$$

$$\text{Up}[B, F] = \text{In}[B, D] @ W_{\text{up}}[D, F] \Rightarrow BF$$

$$\text{Out}[B, D] = \text{Hidden}[B, F] @ W_{\text{down}}[F, D] \Rightarrow BD$$

$$\Rightarrow (2BF + BD)(L) \Rightarrow 2 \cdot 40 \cdot 16e6 \cdot 5120 + 4273 \cdot 16e6 \text{ total activations}$$

$$\text{with } B = 16e6 //$$

3) 32k seq len & total batch size 3M, v5p on 16x16x16.

$$\text{Number of seqs per step} = \frac{3M}{32} = 96 \text{ sequences.}$$

v5p has 96GB HBM per chip.

a) Pure DP - does the full model fit on a chip?

Since pure DP $\Rightarrow$ each chip has full params + full optimizer state and only batch is split, then a chip must be able to hold full params + optim

$$\text{states} = 10 \times 13 \times 10^9 = 130 \times 10^9 = 1306B > 96GB$$

$\Rightarrow$ NO (memory)

b) Pure FSDP: batch split, params and optim also sharded, weights gathered JiT. $\Rightarrow$ is mem ok? comms ok?

Memory: from (Q2), when $B=3M$, Total mem = $\frac{3}{16} \times 42 = 7.9TB$

$$7.9TB + (\text{params} \& \text{optim}) = 7.9TB + 13068 \approx 8TB$$

$$\frac{8TB}{183} \approx 26B \text{ per chip} \Rightarrow \text{memory ok}$$

Comms: The local batch per chip must be big enough

$$\text{where batch per chip} > \frac{2550}{M_x}$$

In this qn, chips = 4096, mesh axes = $M_x = 3$

$$\therefore \text{Required min batch per chip} = \frac{2550}{3} = 850$$

$$\Rightarrow \text{Total batch min} = 850 \times 4096 = 3.48M \text{ tokens}$$

$\therefore 3M < 3.48M$ therefore pure FSDP may not work well here and may be too comms bound

3) Mixed FSDP + TP can drop the required local batch to

$$\frac{2550^2}{2P} = \frac{2550^2}{2(13824)} = 235 \text{ tokens per chip and the actual}$$

$$\text{local batch} = \frac{3M}{4096} = 732 \text{ tokens/chip} \Rightarrow \text{theoretically ok}$$

Step time estimate: Training FLOPs = $6B \times \text{params}$

$$\therefore \text{time} = \frac{6 \times 138 \times 3M}{4096 \times 4.6014 \times 0.4} = 0.3 \text{ seconds}$$

Choosing X & Y s.t. $\frac{BD}{X} \approx \frac{DE}{Y}$ (activation comms $\approx$ weight comms)

$$\frac{B}{X} = \frac{E}{Y} \text{ and } XY = N \Rightarrow Y = \frac{N}{X}$$

$$\therefore \frac{B}{X} = \frac{EX}{N} \Rightarrow X^2 = \frac{BN}{E} \Rightarrow X = \sqrt{\frac{BN}{E}} = \sqrt{\frac{4096(3M)}{13824}} \approx 942$$

$$\therefore \text{Closest } X = 1024, Y = 4$$